

Evidence Summary - Technology #10

Autonomous Catheter Navigation with Human-in-the-Loop

External Evidence Sources (5 total):

Source 1: A Soft-Tip Hydraulically Steerable Catheter for Enhanced

URL: <https://advanced.onlinelibrary.wiley.com/doi/10.1002/aisy.202500099>

Excerpt: This study introduces a hydraulically steerable catheter with a soft tip in vascular procedures. The steering soft tip achieves a minimal ...

Source 2: Full article: Steerable catheters for minimally invasive surgery

URL: <https://www.tandfonline.com/doi/full/10.1080/24699322.2018.1526972>

Excerpt: by X Hu · 2018 · Cited by 217 — The catheters actuated by shape memory effects, conducting polymers and hydraulic pressure require relatively complex fabrication techniques ...

Source 3: Soft robotic steerable micro-catheter for the endovascular

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9809155>

Excerpt: by T Gopesh · 2021 · Cited by 172 — Controlled by hand, the devices offer complete 3D orientation of the tip. Using saline as a working fluid, we demonstrate guidewire-free navigation, access, and ...

Source 4: (PDF) A Soft-Tip Hydraulically Steerable Catheter for

URL: https://www.researchgate.net/publication/392919122_A_Soft-Tip_Hydraulically_Steerable_Catheter_for_Enhanced_Flexibility_and_Safety_in_Vascular_Interventions

Excerpt: Herein, a novel steerable catheter is presented, featuring a hydraulically actuated, soft, steerable tip that allows for real-time visualization ...

Source 5: Development of a smart guide wire using an electrostrictive

URL: <https://www.nature.com/articles/srep18593>

Excerpt: by F Ganet · 2015 · Cited by 62 — Shape memory polymers (SMPs) have polymer and exposing it to an alternating magnetic field, as could be used in conjunction with MRI12.

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*